



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 30/09/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3 rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Statement:- Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements

Objective:- The objective of this practical is to design a MySQL database schema for the StudentHub system. It includes creating tables for **students, events, and registrations** with appropriate relationships. An **ER model** is developed to represent the structure and relationships between the entities. PHP **PDO connectivity** is implemented to establish a connection between the application and MySQL database.

Github Repository Link:-

<https://github.com/Nishtha-cse/StudentHubPortal-D26CS12>

Screenshot

SQL Dump

Server: 127.0.0.1 » Database: studenthub

Structure SQL Search Query Export Import Operations Privileges Routines Events Trigger

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ events	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ registrations	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ students	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
3 tables	Sum	11	InnoDB	utf8mb4_general_ci	80.0 KiB	0 B

☐ Check all With selected:

Server: 127.0.0.1 » Database: studenthub » Table: students

Browse Structure SQL Search Insert Export Import

✓ Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)

```
SELECT * FROM `students`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort

Extra options

	student_id	name	email	course
☐ Edit Copy Delete	1	Nishtha	nishtha@gmail.com	Computer
☐ Edit Copy Delete	2	Anushka	anushka@gmail.com	Computer
☐ Edit Copy Delete	3	Riddhi	riddhi@gmail.com	IT
☐ Edit Copy Delete	4	Neha	neha@gmail.com	Computer

Server: 127.0.0.1 » Database: studenthub » Table: events

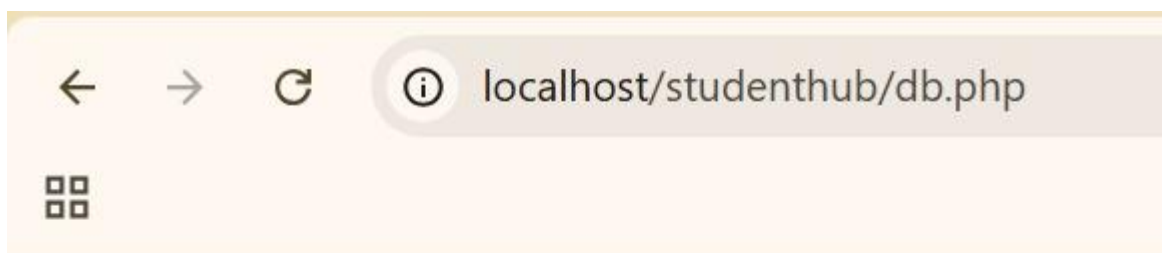
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

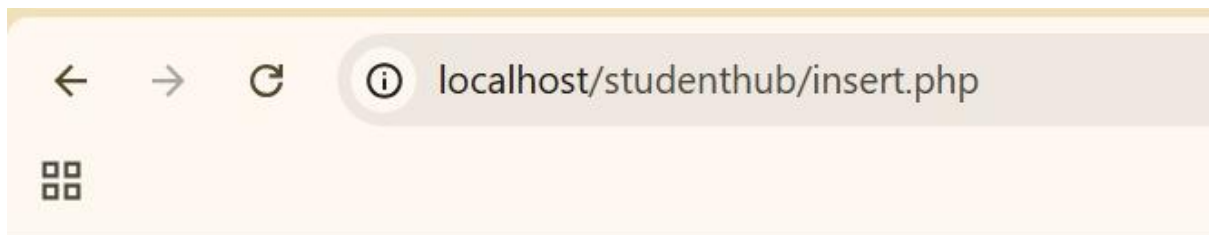
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	event_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	event_name	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 3	event_date	date			Yes	NULL			Change Drop More
☐ 4	venue	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Code:**Db.php**

```
<?php
$host = "localhost";
$dbname = "studenthub";
$username = "root";
$password = "";
try {
    $pdo = new PDO(
        "mysql:host=$host;dbname=$dbname",
        $username,
        $password
    );
    echo "Database Connected
    Successfully";
}
catch (PDOException $e) {
    echo "Connection Failed";
}
?>
```

Screenshot :**Connection screenshot :**

Database Connected Successfully



Learning outcome :**After completing this practical, I will be able to:**

Design a MySQL database schema for the StudentHub system.

Create and manage tables for students, events, and registrations.

Understand and represent relationships between entities using an ER model.

Establish a database connection between PHP and MySQL using PDO.

Use prepared statements to execute SQL queries securely.

Understand the importance of database security and efficient data management in web applications.